THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A solution turns blue litmus paper red. The solution is: 1

(a) salty **(b)** basic **(c)** neutral **(d)** acidic

2. Four cells of 1.5 V each are joined in series. The total voltage is: 1

(a) 6.0 V **(b)** 4.5 V **(c)** 3.0 V **(d)** 1.5 V

3. A substance is shiny, can be hammered into thin sheets and conducts electricity. It is probably a: 1

(a) non-metal **(b)** gas **(c)** metal **(d)** liquid non-metal

4. Which one of these pairs contains only chemical changes? 1

(a) dissolving sugar, freezing water **(b)** cutting paper, tearing cloth

(c) melting wax, boiling water **(d)** burning of coal, ripening of fruit

5. A girl of 13 has a growth spurt before boys of the same age. This is: 1

(a) a disease **(b)** normal **(c)** an illness **(d)** abnormal

6. A rod is heated at one end. The time taken for wax pins to fall off is least for the rod made of: 1

(a) copper **(b)** wood **(c)** glass **(d)** iron

7. A car covers 90 km in 1.5 h. Its speed is: 1

(a) 45 km/h **(b)** 90 km/h **(c)** 60 km/h **(d)** 135 km/h

8. Which of the following correctly shows the order of the alimentary canal? 1

(a) oesophagus, mouth, stomach, small intestine

(b) mouth, oesophagus, stomach, small intestine, large intestine

(c) mouth, stomach, oesophagus, large intestine, small intestine

(d) stomach, mouth, oesophagus, intestine

9. A leaf from a plant kept in the dark and another from sunlight are tested with iodine. The one that turns blue-black is: 1

(a) the dark leaf **(b)** the sunlight leaf

(c) both **(d)** neither

10. A ray of light strikes a plane mirror at an angle of 40° to the normal. The angle between the incident and reflected rays is: 1

(a) 40° **(b)** 50° **(c)** 100° **(d)** 80°

11. If the Earth stopped rotating, the part facing the Sun would: 1

(a) have rain **(b)** have continuous daylight
(c) cool down **(d)** have night

12. In $HCl + NaOH \longrightarrow NaCl + H_2O$, the base is: 1

(a) $NaCl$ **(b)** H_2O **(c)** HCl **(d)** $NaOH$

13. Which of the following is not a safety measure? 1

(a) using insulated tools **(b)** using rubber mats
(c) using MCB **(d)** using damaged wires

14. When magnesium ribbon is burnt and the ash dissolved in water, the solution turns: 1

(a) green **(b)** red litmus blue **(c)** colourless **(d)** blue litmus red

15. In the experiment with three nails, the nail in dry air does not rust because: 1

(a) no oxygen **(b)** no moisture **(c)** no sunlight **(d)** no iron

16. In which of the following are boys and girls same at puberty? 1

(a) growth spurt **(b)** beard growth **(c)** voice deepening **(d)** menstruation

17. During sea breeze, the pressure of air above land is: 1

(a) the same **(b)** higher than over sea
(c) zero **(d)** lower than over sea

18. A car travels 40 km in 1 h and then 60 km in the next hour. Its average speed is: 1

(a) 40 km/h **(b)** 50 km/h **(c)** 60 km/h **(d)** 100 km/h

19. The average number of breaths per minute for a resting adult is 16. The number in an hour is: 1

(a) 160 **(b)** 1600 **(c)** 96 **(d)** 960

20. A potted plant is covered with a polythene bag and kept in sunlight. After some time water droplets are seen inside the bag. This shows: 1

(a) photosynthesis **(b)** digestion **(c)** respiration **(d)** transpiration

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. A solution turns red litmus blue. What does it contain? Give an example. 2

22. A student builds a tester with a cell, bulb and two wires. The bulb glows when a coin is placed but not with a plastic cap. Explain. 2

23. Why do silver articles turn black on keeping in air? 2

24. Why is the burning of a candle called both a physical and a chemical change? 2

25. Why do boys' voices change more than girls' at puberty? 2

26. Why does a metal rod conduct heat faster than a wooden stick? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A solution X turns red cabbage extract green and turmeric red. Another solution Y turns it red and turmeric stays yellow. Identify X and Y. 3

28. Four 1.5 V cells are joined in series to light a 6 V bulb. Does the bulb glow normally? Explain. 3

29. Predict what will happen if a copper nail is dipped in iron sulphate solution and explain. 3

30. You are given a piece of iron nail and a copper sulphate solution. Predict the changes and say whether the change is physical or chemical. 3

31. A teenager is fond of cola and chips and avoids vegetables. Explain the health risks and suggest changes. 3

OR

Explain how an electric heater warms a room using all modes of heat transfer. 3

32. A cyclist covers 12 km in 40 minutes. Find the speed in km/h and m/s. 3

OR

Trace the path of a piece of bread from mouth to anus, with changes at each stage. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. You are given five unlabelled liquids: vinegar, lime water, sugar solution, lemon juice and soap solution. Design a test to identify each. 5

34. Describe a complete experiment to find which materials are conductors, including the circuit diagram in words, table of results and conclusion. 5

35. Plan an investigation to classify ten given materials as metals or non-metals using lustre, hardness, conduction and sound. 5

36. Plan an investigation to find the effect of salt water and plain water on rusting of iron nails, with the set-up, observations and conclusion. 5

OR

Describe the changes at puberty in a table-like form for boys and girls, mentioning the hormones involved. 5

37. Describe the three experiments to demonstrate conduction, convection and radiation. 5

OR

A car starts from city A at 8:00 a.m. and reaches city B 240 km away at 12:00 noon, with a half-hour stop. Find the average speed including the stop and the running speed. 5

38. Explain the whole process of digestion of a meal consisting of chapati (starch), dal (protein) and ghee (fat), naming the organs, juices and the end products. 5

OR

Explain how a leaf is adapted for photosynthesis. 5

END OF QUESTION PAPER — Total: 80 Marks

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